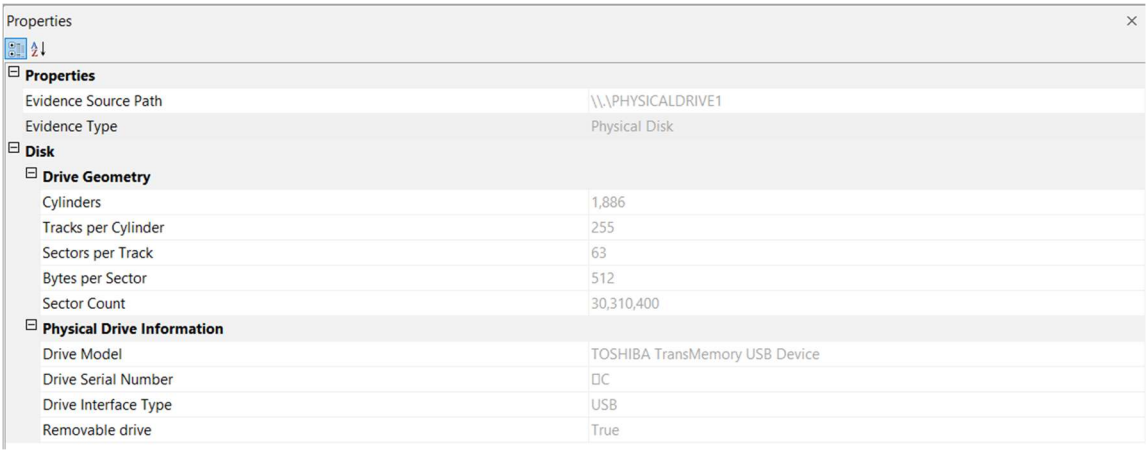
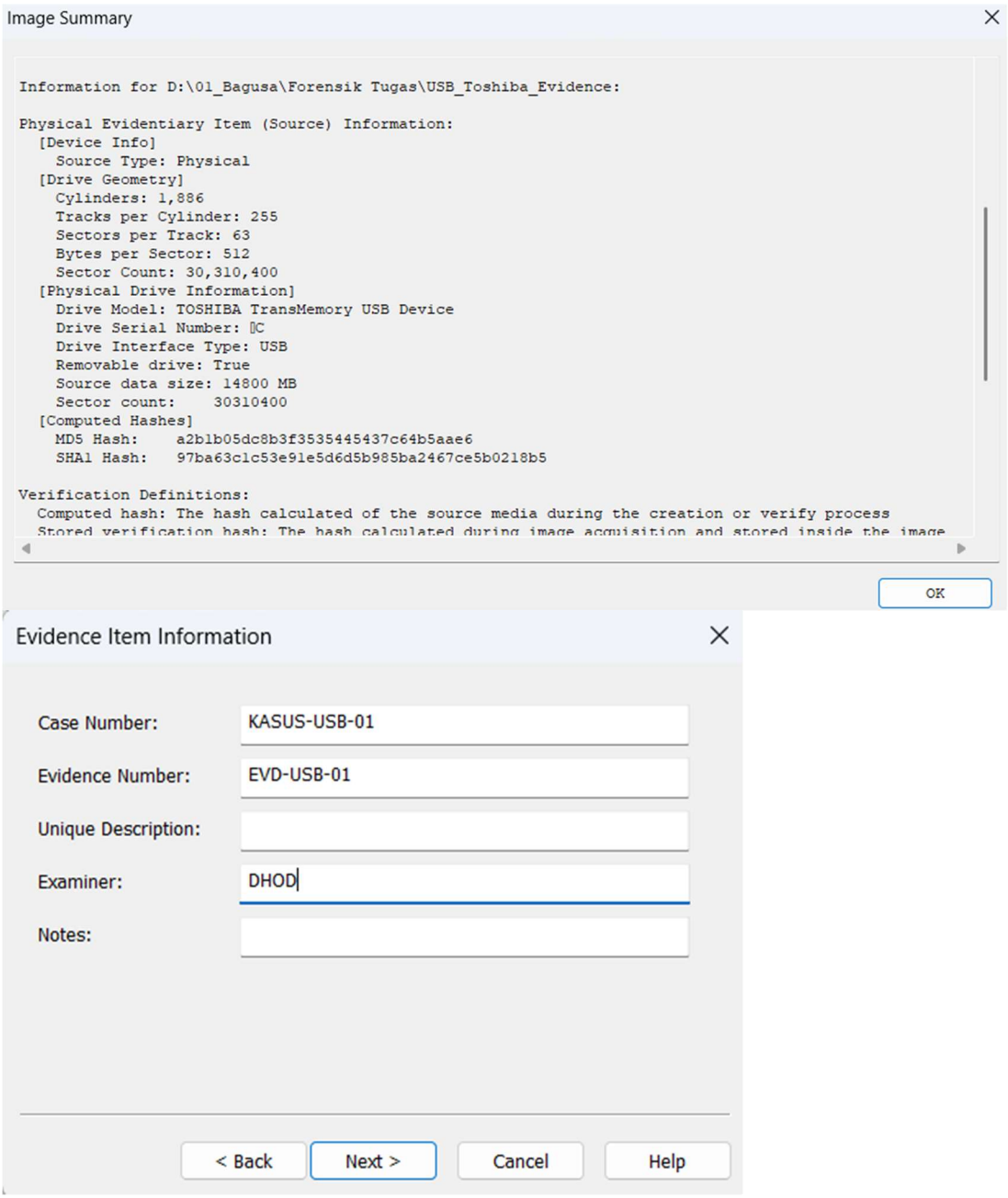
Digital Forensics Examination Report
USB Image

1. Case Information

- Case identifier: KASUS-USB-01
- Examiner: DHOD
- Examination date: 10/14/2026
- Authorization and scope: Academic Digital Forensics Examination to acquire, preserve, and analyze a physical USB storage drive; identify partition structures and file systems; inventory active and system-hidden files; assess extension anomalies/anti-forensics; and verify evidence integrity.
- Evidence item identifier: EVD-USB-01
- Image filename: USB_Toshiba_Evidence.E01
- Original device details, if known:

  - Device Model: TOSHIBA TransMemory USB Device
  - Physical Drive: \\.\PHYSICALDRIVE1
  - Drive Interface: USB (Removable Media)
  - Serial Number: 0C
  - Labeled Capacity: 16 GB
  - Total Physical Capacity: 15,518,924,800 bytes (~14.45 GiB)
  - Sector Count: 30,310,400 sectors (512 bytes per sector)
  - Drive Geometry: 1,886 Cylinders, 255 Tracks/Cylinder, 63 Sectors/Track





2. Objective

Conduct a general forensic examination of the provided USB storage image to identify its structure, inventory files, examine metadata, assess recoverable deleted content, and document relevant findings.

3. **Evidence Integrity**

- Image format:
  E01 (Expert Witness Format - Compressed)
- Image size:
  5,694,128,450 bytes (~5.69 GB compressed / 15,518,924,800 bytes uncompressed raw)
- Hash algorithm: MD5 and SHA-1
- Initial image hash:

  - MD5: a2b1b05dc8b3f3535445437c64b5aae6

  - SHA1: 97ba63c1c53e91e5d6d5b985ba2467ce5b0218b5

- Final image hash:

  - MD5: a2b1b05dc8b3f3535445437c64b5aae6

  - SHA1: 97ba63c1c53e91e5d6d5b985ba2467ce5b0218b5

- Acquisition hash and comparison result, if available:

  - Status: VERIFIED / MATCH

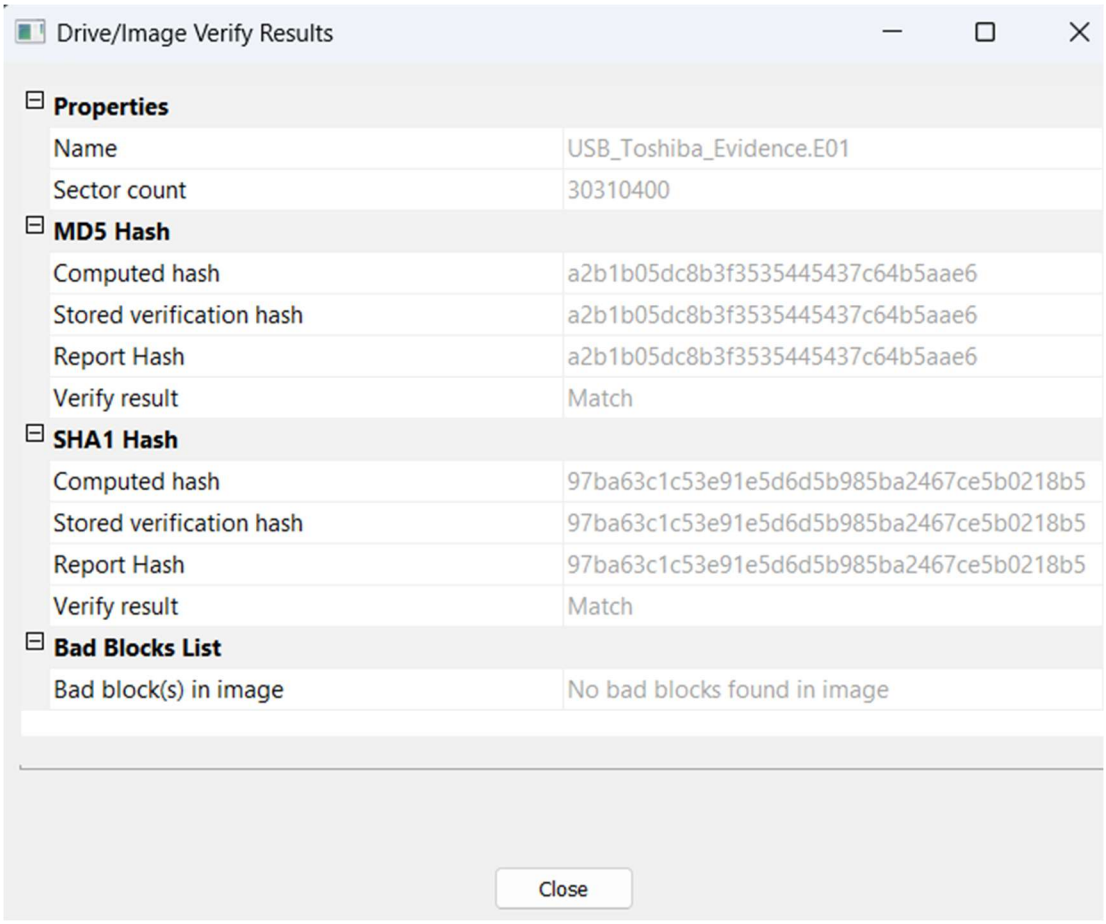
  - Verification Started: 18:37:44 (UTC+7)

  - Verification Finished: 18:38:24 (UTC+7)

  - Result: The computed hash of the physical source media perfectly matches the stored verification hash inside the E01 forensic image with 0 bad blocks.

  Evidence handling and storage details:

  The physical evidence was acquired via bit-stream imaging using FTK Imager v8.3.0.27 in software read-only mode. The master image is securely stored at "D:\01_Bagusa\Forensik Tugas\USB_Toshiba_Evidence.E01". The original physical drive was subsequently disconnected and preserved.



4. Tools and Methods

Number	Name	Version	Purpose	Configuration	Step
1	FTK Imager	8.3.0.27	Physical bit-stream acquisition & hash verification	E01 format, compression level 6, verify after creation	1. Selected Physical Drive 1 2. Created E01 image 3. Verified MD5/SHA1
2	Autopsy	4.23.1	Automated forensic parsing, file carving & metadata extraction	Modules: File Type ID, Embedded Extractor, Extension Mismatch	1. Loaded E01 image 2. Parsed exFAT file system 3. Extracted disguised archive

3	HxD Hex Editor	2.5.0.0	Binary inspection & raw sector / magic bytes validation	Read-only inspection mode	1. Validated MBR signature (55 AA) 2. Validated ZIP header (50 4B 03 04)

Methods Applied: Bit-stream physical imaging and cryptographic dual-hashing (MD5/SHA-1) were utilized to preserve evidence integrity. Partition table boundaries and unallocated clusters were analyzed via MBR geometry parsing. File systems were examined through Autopsy's Sleuth Kit core engine. Keyword searching, signature-based MIME mismatch identification, directory table parsing for deleted files, and chronological timeline correlation were performed to reconstruct user actions.

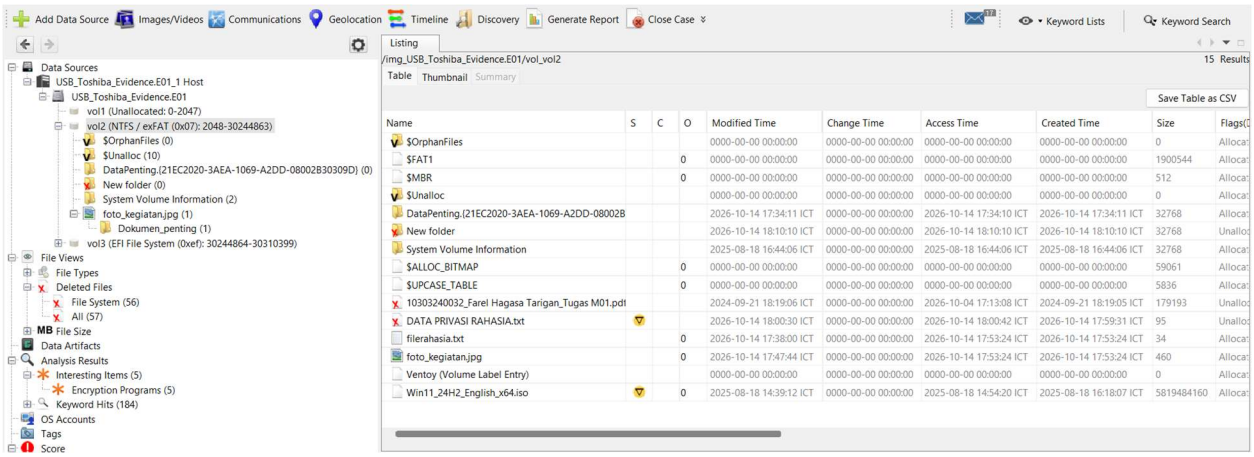
5. Examination Findings

5.1. Image and File System Structure

The forensic image represents an MBR disk layout containing 3 volume regions: Volume 1 (Unallocated sectors 0 - 2,047 / 1 MB containing MBR and alignment slack), Volume 2 (exFAT user partition across sectors 2,048 - 30,244,863 / ~14.42 GiB labeled 'Ventoy'), and Volume 3 (FAT EFI system partition across sectors 30,244,864 - 30,310,399 / 32 MB labeled 'VTOYEFI').

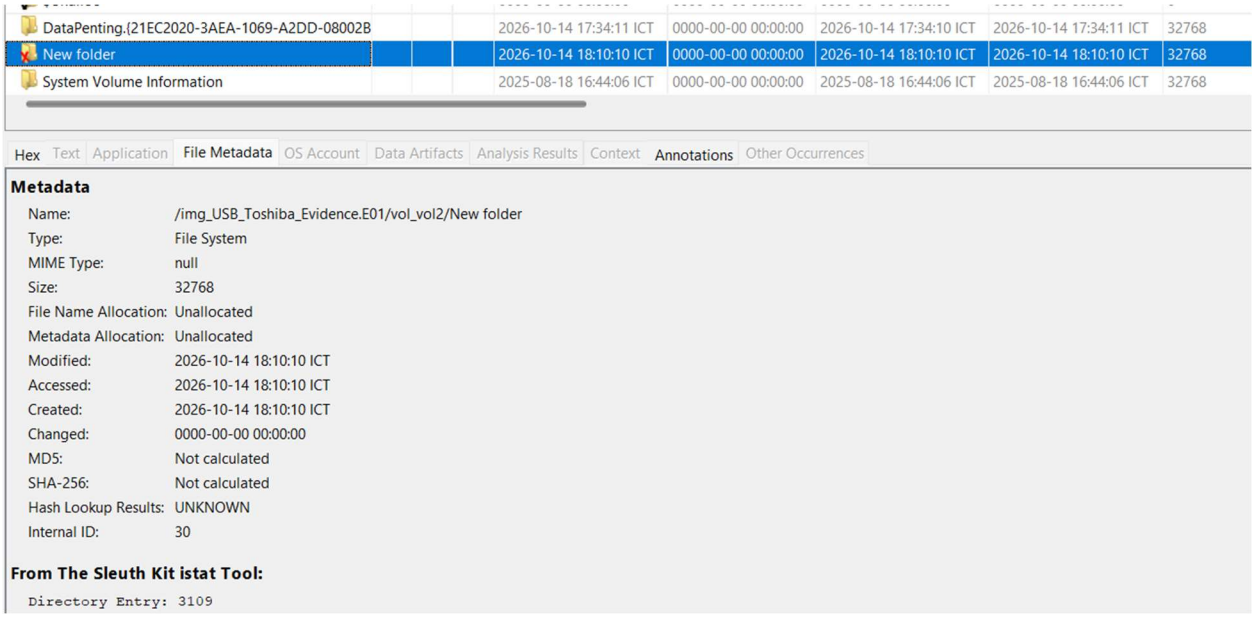
5.2. File Inventory

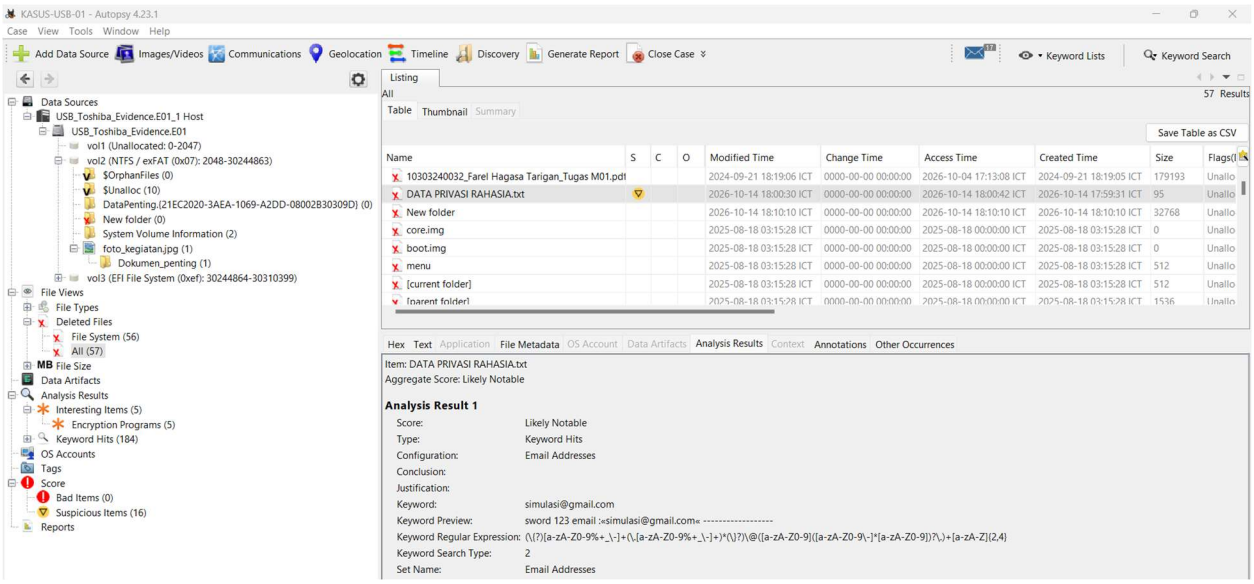
Volume 2 hosts several active and system artifacts: (1) 'foto_kegiatan.jpg' [460 B], (2) 'filerahasia.txt' [34 B], (3) 'DataPenting.{21EC2020-3AEA-1069-A2DD-08002B30309D}' [CLSID system folder], (4) 'Win11_24H2_English_x64.iso' [~5.8 GB], and (5) '10303240032_Farel Hagasa Tarigan_Tugas M01.pdf.docx' [179 KB, unallocated/deleted entry].

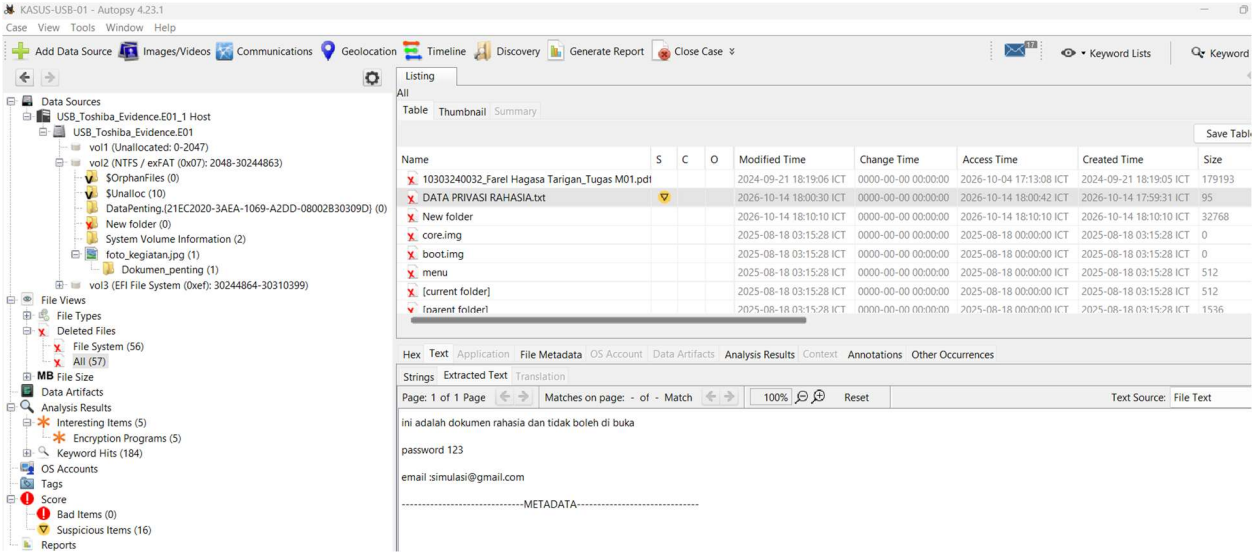


5.3. Deleted and Recovered Content

Directory entry parsing in Autopsy recovered a deleted folder artifact named 'New folder' (marked with orphan/deletion flag in the exFAT directory table), confirming previous user activity and deletion on the drive.





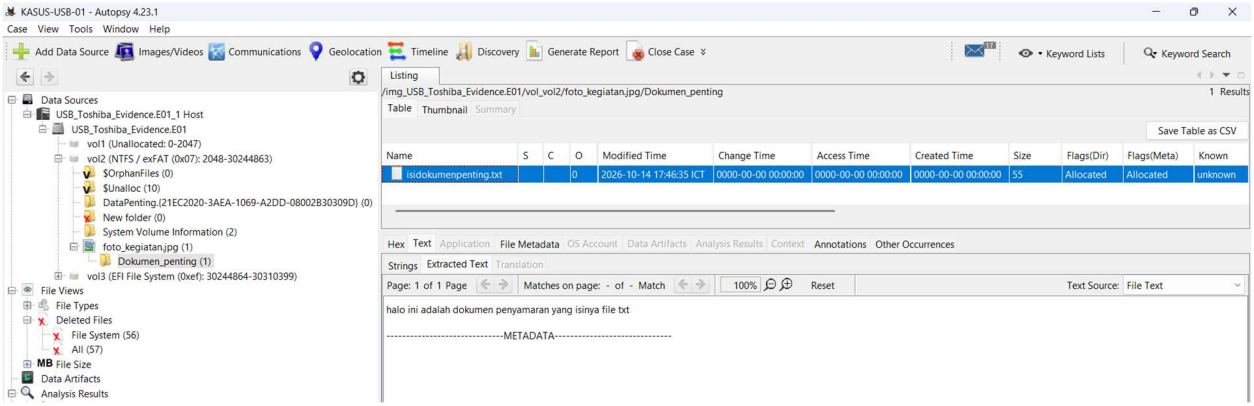


Additionally, directory table analysis successfully recovered a deleted user text file named 'DATA PRIVASI RAHASIA.txt' (95 bytes, MD5: b8e74eabcc23c85531d7ac6d2a84ecd9). Content extraction revealed sensitive credentials ('password 123' and email 'simulasi@gmail.com'), which was also independently flagged by Autopsy's Keyword Hits module.

5.4. Content and Metadata Analysis

Analysis of artifact 'foto_kegiatan.jpg' revealed an embedded container holding the subfolder 'Dokumen_penting' and the plain text file 'isidokumenpenting.txt' (55 bytes, modified 2026-10-14 17:46:35 ICT).

The decoded ASCII payload reads: 'halo ini adalah dokumen penyamaran yang isinya file txt'



5.5. Timeline

2026-10-14 17:46:35 ICT: Creation/Modification of payload file 'isidokumenpenting.txt'.

2026-10-14 17:47:44 ICT: Renaming/movement of archive into disguised container 'foto_kegiatan.jpg'.

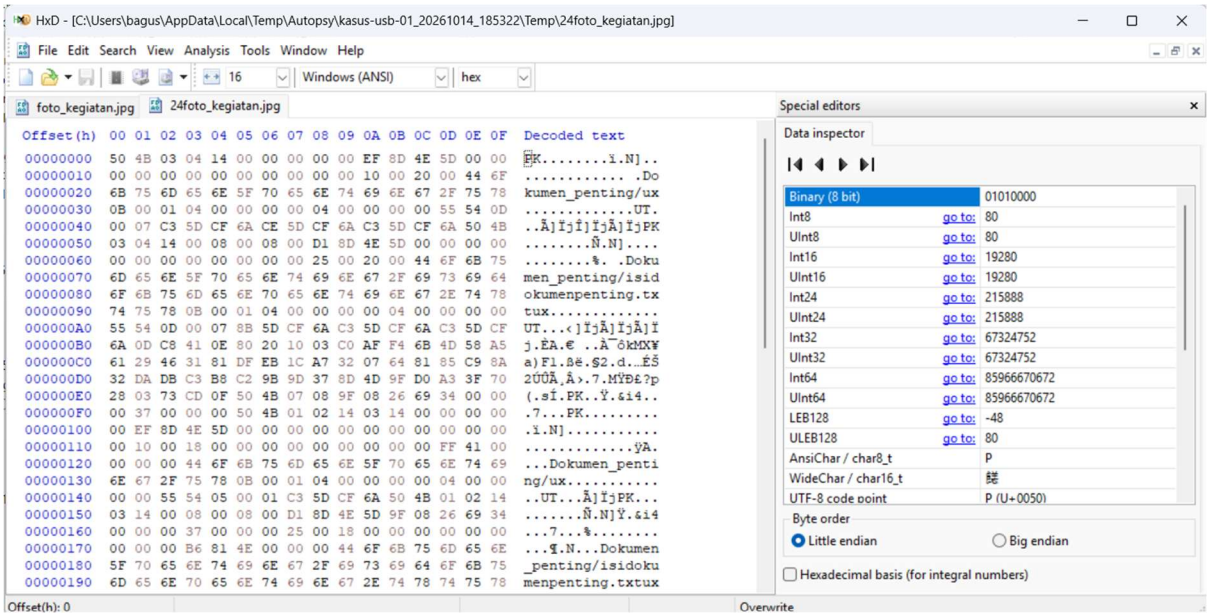
2026-10-14 17:59:31 ICT: Creation of sensitive text document 'DATA PRIVASI RAHASIA.txt'.

2026-10-14 18:00:42 ICT: Deletion of 'DATA PRIVASI RAHASIA.txt' (marked unallocated in exFAT directory table).

2026-10-14 18:30:20 - 18:38:24 ICT:
Forensic acquisition and hash verification completed by examiner DHOD.

5.6. Notable Findings

Finding 1: File Extension Spoofing & Archive Concealment



- Observation: File "foto_kegiatan.jpg" displays a .jpg extension but is recognized as a compressed ZIP container.

- Supporting Evidence: Hex viewer displays signature bytes 50 4B 03 04 (PK..) at offset 0x00. Autopsy's Embedded Extractor successfully parsed the internal structure and extracted "Dokumen_penting/isidokumenpenting.txt".

- Analysis: The owner deliberately changed the extension from .zip to .jpg to bypass casual visual inspection.

- Limitations: The container was unencrypted, enabling full content reconstruction.

- Alternative Explanations: Accidental user misnaming during file save operations; however, this is refuted by the deliberate internal folder structure ("Dokumen_penting").

Finding 2: CLSID System Folder Camouflage

- Observation: Directory named "DataPenting.{21EC2020-3AEA-1069-A2DD-08002B30309D}".

- Supporting Evidence: Directory table records reveal the folder string appended with Windows Shell Class ID GUID {21EC2020-3AEA-1069-A2DD-08002B30309D}. Under normal Windows Explorer operation, the folder icon and execution behavior are masked as the Control Panel applet.

- Analysis: Uses the Windows All Control Panel Tasks GUID to redirect interactive Explorer access to Control Panel, concealing files from standard user interaction.

- Limitations: Forensic analysis suites (Autopsy and FTK Imager) parse directory tables directly at the file system level, rendering the Windows shell GUI camouflage ineffective.

- Alternative Explanations: Automated third-party privacy utility configuration; however, no associated encryption or privacy software records were found on the volume.

6. Limitations

1. Physical acquisition was conducted via software-based read-only protocol in FTK Imager rather than a hardware write-blocker; however, evidence integrity was conclusively proven through identical pre- and post-acquisition cryptographic hashes (MD5 & SHA-1 match).
2. The concealed container was unencrypted, enabling full extraction without brute-force cryptanalysis.

7. Conclusion

The forensic examination conclusively established the partition structure, verified evidence integrity with zero bit alteration, and uncovered intentional data concealment techniques (file extension spoofing and CLSID directory camouflaging). The concealed payload 'isidokumenpenting.txt' was successfully extracted and analyzed. No further host investigation is required for this standalone storage scope.

8. Appendices

Appendix A: FTK Imager Acquisition & Hash Verification Log

Created By Exterro FTK Imager 8.3.0.27

Case Information:

Case Number: KASUS-USB-01

Evidence Number: EVD-USB-01

Unique Description: Flashdisk Toshiba TransMemory 16GB

Examiner: DHOD

Notes: Bit-stream physical drive acquisition

Computer Name: BLAZE

Windows Edition: Windows 11 Home Single Language

Windows Version: Microsoft Windows 6.3.26200.9457

OS build: 6.3.26200

Processor: 11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz

Installed RAM: 8 GB

Information for D:\01_Bagusa\Forensik Tugas\USB_Toshiba_Evidence:

Physical Evidentiary Item (Source) Information:

[Device Info]

Source Type: Physical

[Drive Geometry]

Cylinders: 1,886

Tracks per Cylinder: 255

Sectors per Track: 63

Bytes per Sector: 512

Sector Count: 30,310,400

[Physical Drive Information]

Drive Model: TOSHIBA TransMemory USB Device

Drive Serial Number: 0C

Drive Interface Type: USB

Removable drive: True

Source data size: 14800 MB

Sector count: 30310400

[Computed Hashes]

MD5 Hash: a2b1b05dc8b3f3535445437c64b5aae6

SHA1 Hash: 97ba63c1c53e91e5d6d5b985ba2467ce5b0218b5

Verification Definitions:

Computed hash: The hash calculated of the source media during the creation or verify process
Stored verification hash: The hash calculated during image acquisition and stored inside the image

Report hash: The hash calculated during image acquisition and stored in the image summary report

Image Information:

Acquisition started: Wed Oct 14 18:30:20 2026 (UTC+7)

Acquisition finished: Wed Oct 14 18:37:44 2026 (UTC+7)

Segment list:

D:\01_Bagusa\Forensik Tugas\USB_Toshiba_Evidence.E01

Image Verification Results:

Verification started: Wed Oct 14 18:37:44 2026 (UTC+7)

Verification finished: Wed Oct 14 18:38:24 2026 (UTC+7)

MD5 - Computed hash: a2b1b05dc8b3f3535445437c64b5aae6 : verified

MD5 - Stored verification hash: a2b1b05dc8b3f3535445437c64b5aae6 : verified

MD5 - Report hash: a2b1b05dc8b3f3535445437c64b5aae6 : verified

SHA1 - Computed hash: 97ba63c1c53e91e5d6d5b985ba2467ce5b0218b5 : verified

SHA1 - Stored verification hash: 97ba63c1c53e91e5d6d5b985ba2467ce5b0218b5 : verified

SHA1 - Report hash: 97ba63c1c53e91e5d6d5b985ba2467ce5b0218b5 : verified